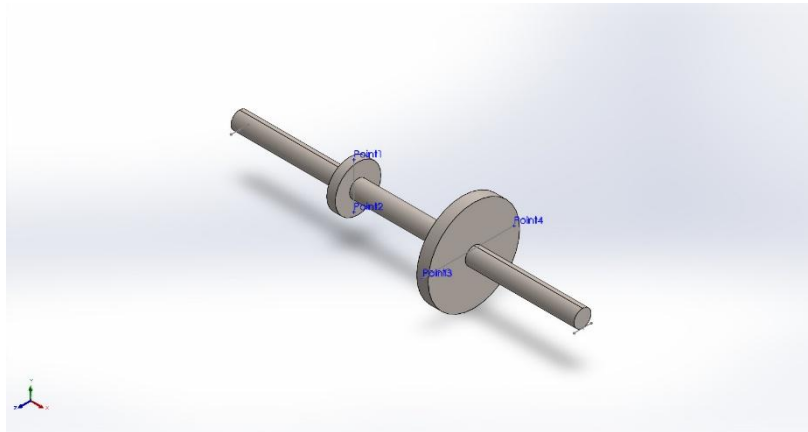




Description

Geometry of the Shaft

Shape: Solid circular rod.

Diameter: 1.5 in.

Length: 40 in (10 in + 10 in + 10 in + 10 in).

Pulleys:

Pulley B: Diameter = 4.0 in, located 10 in from support A.

Pulley C: Diameter = 8.0 in, located 20 in from support A.

Simulation of Part2

Date: Friday, February 13, 2026

Designer: MD Naiem Gazi

Study name:

Shaft_1pt5in_Torsional_Bending_Stress_Study

Analysis type: Static

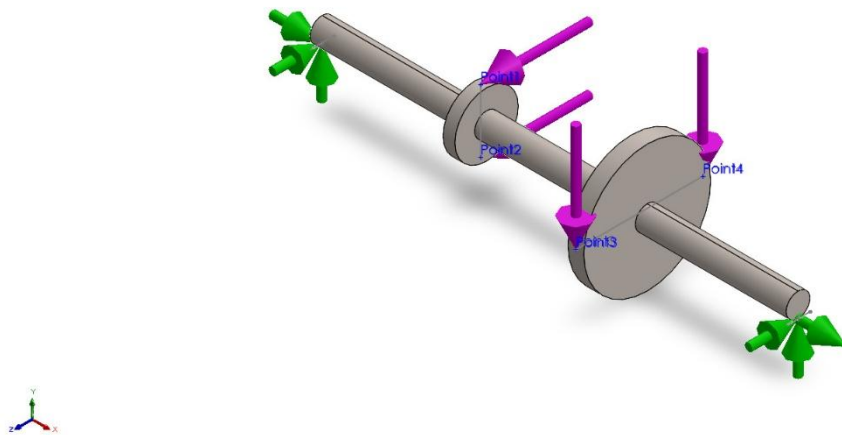
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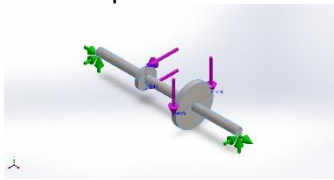
Assumptions

Model Information



Model name: Part2
Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Split Line10 	Solid Body	Mass:13.8309 kg Volume:0.00179622 m ³ Density:7,700 kg/m ³ Weight:135.543 N	E:\L-3 T- 2\Assignments\Part 2\Part2.SLDPRT Jan 29 14:06:36 2023

Study Properties

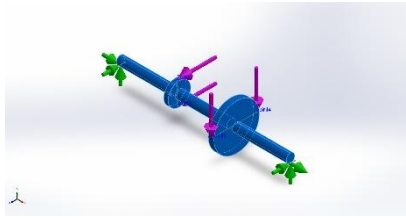
Study name	Shaft_1pt5in_Torsional_Bending_Stress_Study
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Large displacement	On
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (E:\L-3 T-2\Assignments\Part 2)

Units

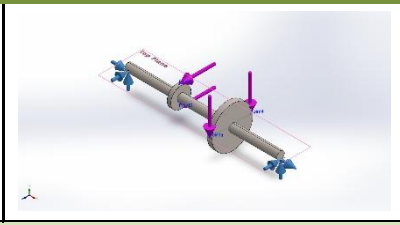
Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

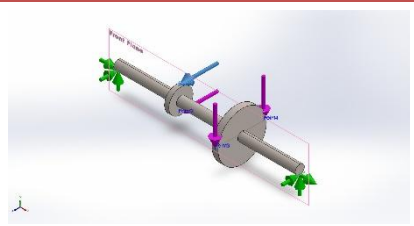
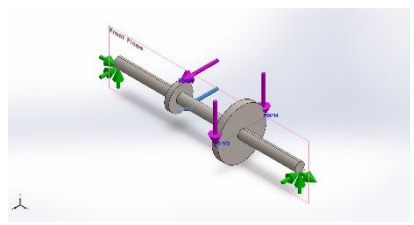
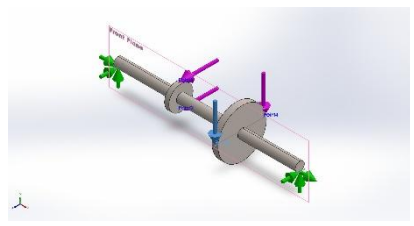
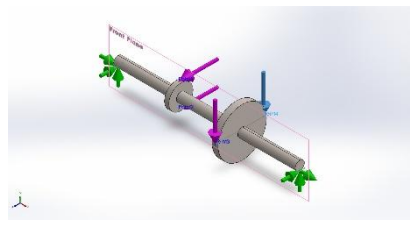


Material Properties

Model Reference	Properties	Components
	Name: Alloy Steel Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 6.20422e+08 N/m² Tensile strength: 7.23826e+08 N/m² Elastic modulus: 2.1e+11 N/m² Poisson's ratio: 0.28 Mass density: 7,700 kg/m³ Shear modulus: 7.9e+10 N/m² Thermal expansion coefficient: 1.3e-05 /Kelvin	SolidBody 1(Split Line10)(Part2)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details		
Reference Geometry-1		Entities: 2 vertex(s), 1 plane(s) Reference: Top Plane Type: Use reference geometry Translation: 0, 0, 0 Units: ft		
Resultant Forces				
Components	X	Y	Z	Resultant
Reaction force(N)	-1.52588e-05	26.6893	-53.3787	59.6791
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Force-1		Entities: 1 plane(s) Reference: Front Plane Type: Apply force Values: ---, ---, 2 lbf
Force-2		Entities: 1 plane(s) Reference: Front Plane Type: Apply force Values: ---, ---, 10 lbf
Force-3		Entities: 1 plane(s) Reference: Front Plane Type: Apply force Values: ---, -5, --- lbf
Force-4		Entities: 1 plane(s) Reference: Front Plane Type: Apply force Values: ---, -1, --- lbf

Connector Definitions

No Data

Interaction Information

No Data

Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	0.957358 in
Minimum element size	0.957358 in
Mesh Quality	High

Mesh information - Details

Total Nodes	2565
Total Elements	1275
Maximum Aspect Ratio	7.068
% of elements with Aspect Ratio < 3	96.2
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:03
Computer name:	HELAL



Sensor Details

No Data

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-1.52588e-05	26.6893	-53.3787	59.6791

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0	0	0	0

Free body moments

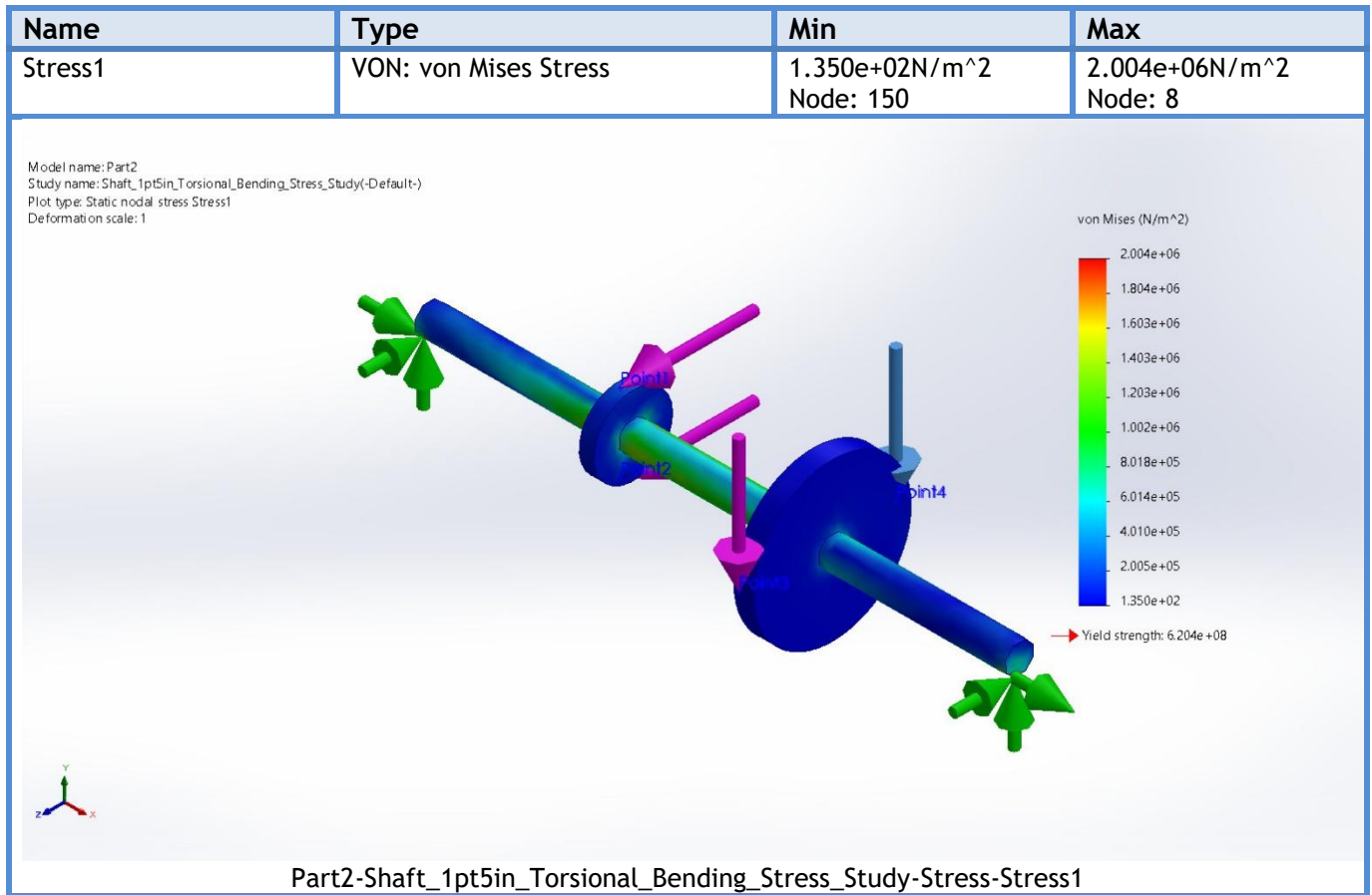
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Beams

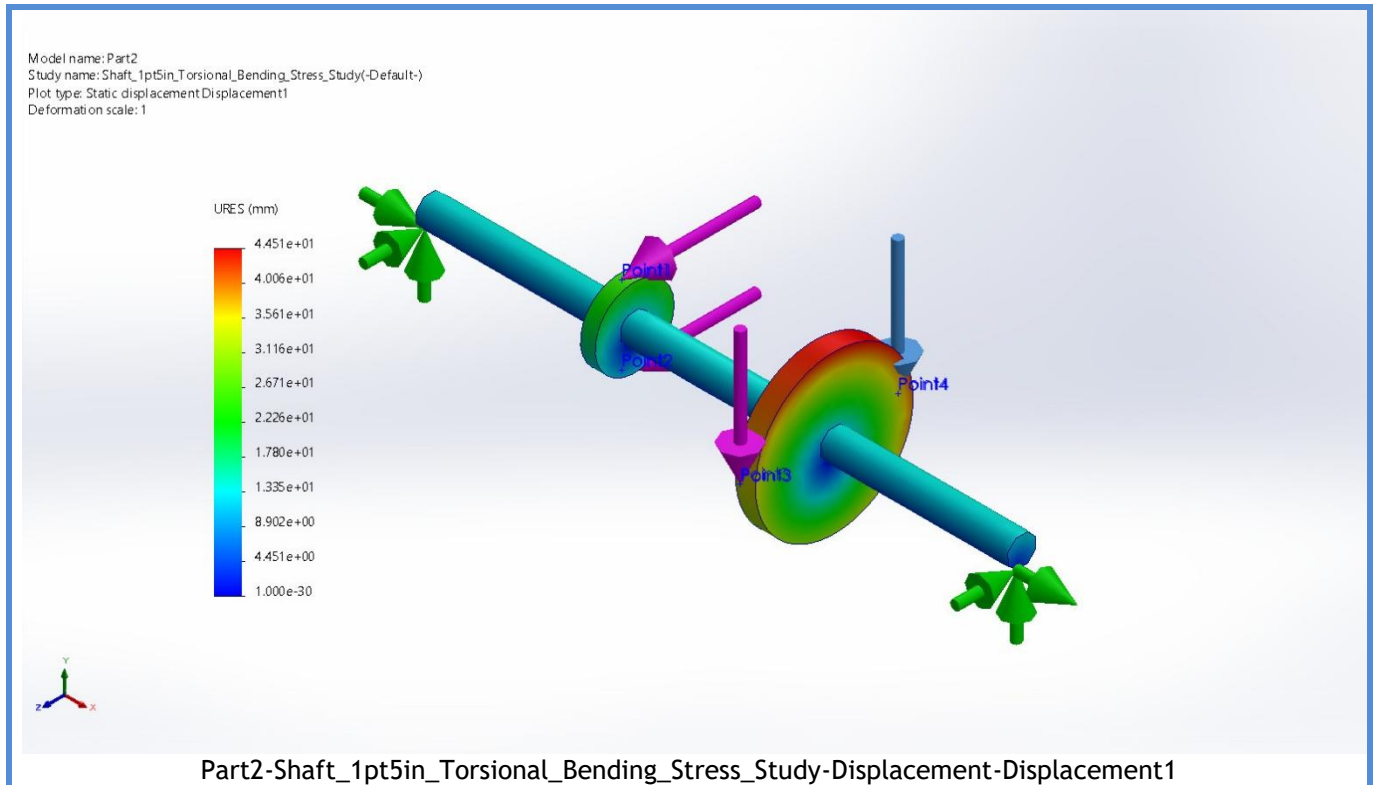
No Data



Study Results



Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 2	4.451e+01mm Node: 131



Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	2.392e-09 Element: 100	4.970e-06 Element: 749

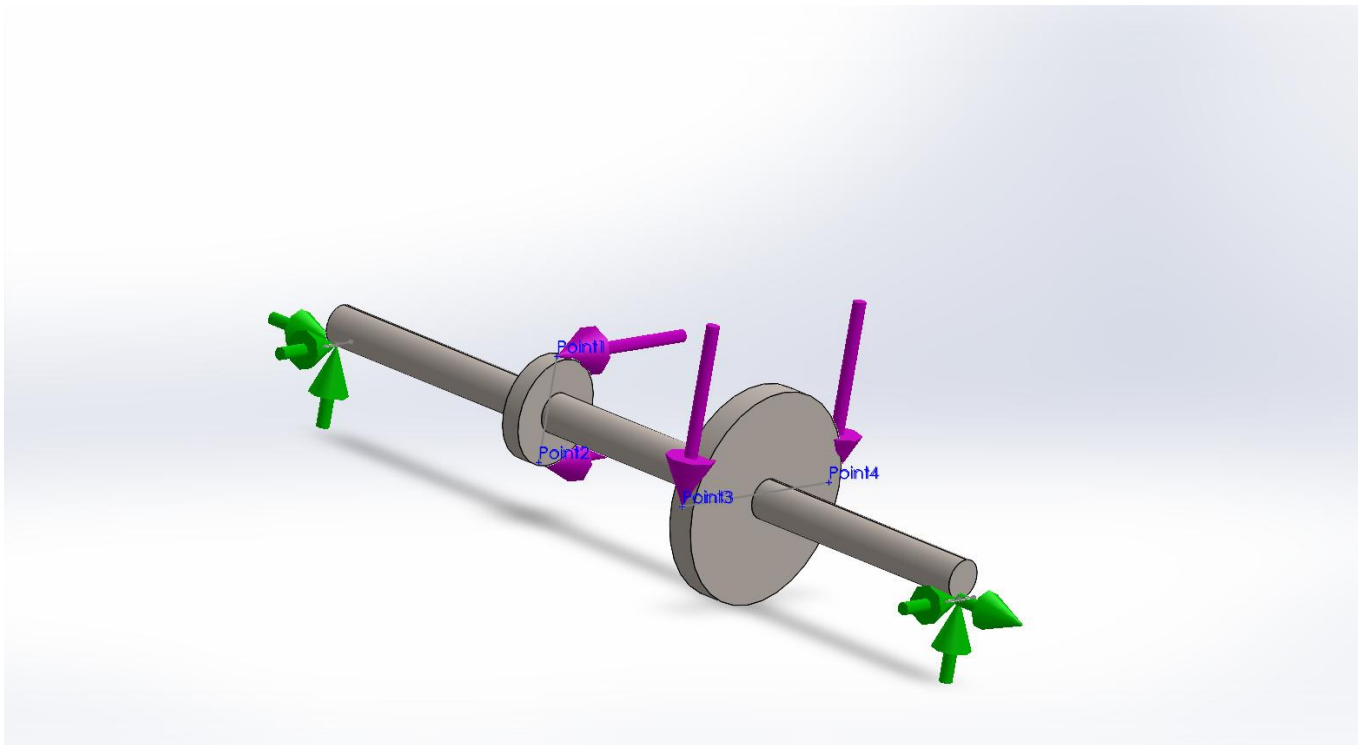
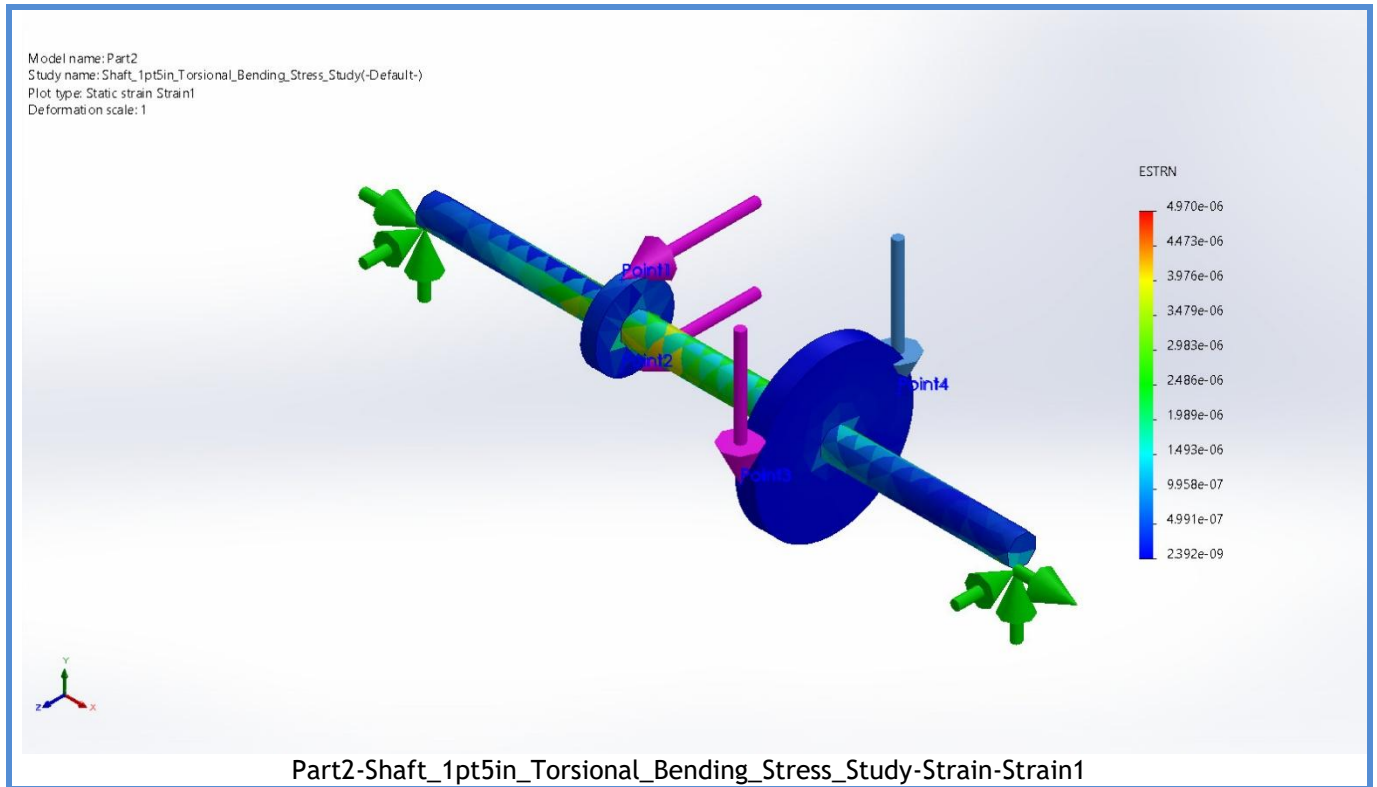


Image-1

Conclusion

